

Combinatorics PhD Exam, May 2018

1. Consider the set T_n of (not rooted) trees with $n \geq 3$ labeled leaves for which each interior (non-leaf) vertex has degree 3.

- (a) Prove that each tree in T_n has exactly $2n - 3$ edges.
- (b) Prove that the number of trees in T_n is $(2n - 5)!! = 1 \cdot 3 \cdot 5 \cdot 7 \cdots (2n - 7) \cdot (2n - 5)$.

2. **Warning: Suspected error.** As printed, part (a) does not yield the Erdős–Szekeres theorem stated in part (b): setting $r = s = n$ gives only size n , while setting $r = s = n + 1$ does not satisfy the stated threshold. The intended correction is not unique, so the source reading has been retained.

The famous Dilworth Theorem for a finite poset P states that the minimum number of chains in any partition of P into chains is equal to the maximum number of elements in an antichain of P .

- (a) Prove that if P has size at least $rs + 1$, then there is either a chain of size r or an antichain of size s .
- (b) There are many proofs of the following Erdős–Szekeres theorem: For any sequence $a_1, a_2, \dots, a_{n^2+1}$ of integers, there is a subsequence of length $n + 1$ that is monotone. Prove the Erdős–Szekeres theorem using Dilworth’s theorem, i.e., part (a).

3. Prove that

$$\sum_{i=0}^n (-1)^i \binom{n}{i} i^k = \begin{cases} 0 & \text{if } 0 \leq k < n, \\ (-1)^n n! & \text{if } k = n. \end{cases}$$

What is the sum if $k > n$?

Hint: Count surjective mappings from $[k]$ to $[n]$.

4. Recall: The girth of a graph is the length of the shortest cycle (and ∞ if the graph has no cycles).
- (a) Prove that a planar graph of girth (minimum cycle length) at least 6 has a vertex of degree at most 2.
 - (b) The Four Color Theorem states that any planar graph is 4-colorable. Grötzsch proved that any triangle-free planar graph is 3-colorable. Without using his result, prove that a planar graph of girth at least 6 is 3-colorable.

5. **Warning: Suspected error.** The problem does not specify the length of the strings, so the requested finite count is underdetermined. A likely missing condition is a fixed length, but the intended wording cannot be determined uniquely from the source.

How many strings can be formed using the alphabet $\{A, B, C, D, E\}$ if

- (a) the letter A occurs an odd number of times
- (b) the letters A and B are both used an odd number of times.

6. Let $f(n)$ be the number of ways to tile a $1 \times n$ path with 1×2 tiles that are red, blue, or green, and 1×1 tiles that are yellow, orange, black, or white.

- (a) Find an explicit formula for $f(n)$.
- (b) On average, how many 1×1 tiles will a $1 \times n$ path contain?

7. Choose a derangement p of length n uniformly at random. Recall that a derangement is a permutation with no 1-cycles. On average, how many 2-cycles does p contain?
8. Let C be a binary code of length n and minimum distance $d \geq 2e + 1$. Prove the following two bounds (the Hamming bound and Singleton bound).

$$|C| \leq \frac{2^n}{\sum_{i=0}^e \binom{n}{i}}$$

$$|C| \leq 2^{n-d+1}$$